The relative importance of body size, phylogeny, locomotion, and diet in the evolution of forelimb dexterity in fissiped carnivores (Carnivora)

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Abstract: Using a new rating scale of forelimb dexterity that separates the contribution of proximal components (shoulder, upper forelimb, and lower forelimb) from distal components (forepaw), we examined the relationship between functional demands and phylogeny and forelimb dexterity in 45 species of fissiped carnivores (Carnivora). Specifically, we examined the effects of body size, phylogenetic relatedness, diet (vertebrate predation), and locomotion (arboreality) on the differential evolution of forelimb dexterity. Regression analyses indicate that, although body size does appear to be positively correlated with the dexterity of the proximal components, the inclusion of phylogenetic information results in a nonsignificant relationship. Phylogenetic relatedness was found to account for a significant amount of interspecific variation in proximal, distal, and total (proximal + distal) dexterity. When phylogenetic effects were incorporated, arboreality was not significantly correlated with any of the dexterity scores, but vertebrate predation was, albeit a negative correlation. The amount of variation in the dexterity of proximal and distal components did, however, differ in magnitude within each significant result. Thus, each component can be differentially affected by specific functional demands. By examining the significant associations with diet and phylogeny and mapping the dexterity scores onto the phylogeny, we also demonstrate that the ancestral degree of forelimb dexterity of both the caniform and feliform lineages was approximately equal to that of the average extant carnivore. Thus, forelimb dexterity has decreased or increased within particular lineages, with reductions or elaborations in some species resulting from the invasion of specific niches not occupied by congeners.

Résumé : Selon une nouvelle échelle propre à mesurer la dextérité des membres antérieurs en séparant la contribution des composantes proximales (épaules, avant-bras et bras) de celle des composantes distales (patte antérieure), nous avons examiné la relation entre les besoins fonctionnels et la dextérité du membre antérieur chez 45 espèces de carnivores fissipèdes (Carnivora). Plus spécifiquement, nous avons examiné les effets de la taille du corps, des relations phylogénétiques, du régime alimentaire (prédation de vertébrés) et de la locomotion (déplacements dans les arbres) sur l'évolution différentielle de la dextérité du membre antérieur. Les analyses de régression indiquent que même s'il semble y avoir une corrélation positive entre la taille du corps et la dextérité des composantes proximales, si on tient compte de l'information phylogénétique la relation n'est plus significative. La parenté phylogénétique explique une grande partie de la variation interspécifique de la dextérité proximale, de la dextérité distale et de la dextérité totale (proximale + distale). En incorporant aux analyses les effets phylogénétiques, la capacité de se déplacer dans les arbres n'est en corrélation avec aucun des facteurs de dextérité, mais la prédation de vertébrés l'est, bien qu'il s'agisse d'une relation négative. L'importance de la variation dans la dextérité des composantes proximales et distales diffère d'amplitude pour chaque résultat significatif. De cette façon, chaque composante peut être affectée différemment par des besoins fonctionnels spécifiques. En examinant les associations positives avec le régime alimentaire et la phylogénie et en cartographiant les résultats de la dextérité en fonction de la phylogénie, nous démontrons que le degré ancestral de dextérité du membre antérieur chez les deux lignées, caniforme et féliforme, est à peu près égal à celui d'un carnivore actuel moyen. Donc, la dextérité du membre antérieur a diminué ou augmenté chez des lignées en particulier, par réduction ou développement chez certaines espèces, résultant de l'invasion de niches spécifiques non occupées par des congénères.

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Introduction

Most studies of forelimb dexterity in mammals focus only upon the contribution of the forepaws and digits (Napier 1961, 1980; Heffner and Masterton 1975, 1983). The problem with this approach is that it ignores the potential contribution of postural and upper forelimb movements to the execution of reaching, grasping, and manipulation. Recently, it

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has been shown that this contribution can form a significant part of an animal's dexterity so that the functional ability to manipulate items with the forepaws can arise through a combination of postural adjustments and movements of all limb segments (Whishaw and Pellis 1990; Iwaniuk et al. 1998; Iwaniuk and Whishaw 1999). Therefore, forelimb dexterity should not be considered to be a property exclusive to the forepaws, and the forelimb cannot be considered independently of the body.

An alternative approach is to consider forelimb dexterity to be a result of the movements of distinct components that act in concert as a functional unit. In other words, the forelimb can be separated, both anatomically and behaviourally, into components that combine to produce reaching, grasping, and manipulative movements. A similar approach has been adopted to investigate the evolution of avian flight (Gatesy and Dial 1996). In their study, Gatesy and Dial (1996) considered the hindlimbs, tail, and wings of birds to be "modules" that can be elaborated upon individually, or in combination with one another, to produce various phenotypes with an array of flying skills. Using their approach as a model, we can then consider the proximal (shoulder, upper forelimb, and lower forelimb) and distal (forepaw) parts of the forelimb as separate components (i.e., modules) that can be elaborated upon individually, or together, to produce a variety of manipulative skills. The development of a new dexterity index (Iwaniuk et al. 1999) that incorporates the contribution of both proximal and distal components to forelimb functioning allows for the separate examination of both components.

Given that the proximal and distal components are organized differently and can, potentially, be elaborated upon separately from one another, one would expect that the functional demands of different morphologies, phylogenetic histories, and behaviours may be differentially related to the elaboration of each component. For example, it is well known that overall body size plays a significant role in the evolution of behavioural and ecological diversity (Schmidt-Nielsen 1984; LaBarbera 1989; Damuth and MacFadden 1990). Since the shape and function of individual elements of the forelimb can be affected by changes in body size (Biewener 1989), one might expect that body size may constrain each component differently.

Similarly, if the ancestors of a species possessed relatively poor proximal forelimb dexterity, it may be unlikely for extant species to have a high degree of proximal dexterity. They may, however, be capable of elaborating upon their distal dexterity, resulting in an overall increase in manipulative prowess. Thus, phylogenetic relatedness may determine not only the degree of forelimb dexterity of a species, but also where the elaboration(s) are likely to occur.

The most frequently cited reasons for interspecific differences in forelimb dexterity are, however, the behavioural demands of different lifestyles (Cartmill 1985; Gittleman 1986*a*; Whishaw et al. 1998). Both locomotion and diet have been previously cited as causally related to differences in forelimb dexterity because of the functional demands placed upon the forelimbs during both activities. For example, arboreal locomotion has long been associated with high degrees of forelimb dexterity (Cartmill 1985). It has been assumed that movement through the trees requires grasping forepaws and this grasping ability is then transferred into nonlocomotory activities. Grasping only involves the forepaws, whereas arboreal locomotion requires the active involvement of the entire forelimb. Therefore, arboreal locomotion may not necessarily be correlated with the dexterity of both forelimb components.

A corresponding argument has also been proposed for a correlation between vertebrate predation and forelimb dexterity (Hopson 1977; Gittleman 1986*a*). The capture of rapidly moving prey is thought to require extensive manipulation and rapid use of the forelimbs (Hopson 1977; Gittleman 1986*a*). Predation and manipulation may, however, require differential use of the proximal parts of the forelimbs as opposed to the forepaws. Thus, vertebrate predation may also affect the dexterity of the distal components separately from that of the proximal components.

Lastly, it may be instructive to determine the likely forelimb dexterity of ancestral species. The evolution from this ancestral state may be determined by the functional demands outlined above, but could also provide some insight into the selective forces responsible for the observed variation in forelimb dexterity. The pattern of changes in forelimb dexterity across lineages may also suggest alternative explanations for observed behavioural diversity.

Extant carnivores have been frequently used to test a variety of ecological and anatomical hypotheses (Gittleman and Harvey 1982; Gittleman 1986*a*, 1986*b*, 1991; Gittleman and Van Valkenburgh 1997). Their wide range of body sizes and locomotor and dietary specializations makes them an ideal group within which to examine the evolution of forelimb dexterity with respect to the different selective forces outlined above (Ewer 1973; Wyss and Flynn 1993). In addition, they are known to exhibit a wide range of forelimb dexterity, ranging from the manipulative capabilities of the giant panda (*Ailuropoda melanoleuca*) (Davis 1967; Kleiman 1983) to the lack of manipulative forelimb use in the cheetah (*Acinonyx jubatus*) (Eaton 1972; Van Valkenburgh 1996). Therefore, we filmed 45 carnivore species and assessed their forelimb dexterity using the index of Iwaniuk et al. (1999). Based upon the theories presented above, the following relationships were predicted: (1) body size is negatively correlated with forelimb dexterity, (2) phylogeny plays a significant role in the variability in forelimb dexterity, (3) arboreality is positively correlated with forelimb dexterity, and (4) vertebrate predation is positively correlated with forelimb dexterity. These four predictions were then tested through ANOVAs of species as independent data points and modern comparative methods (i.e., phylogenetic autocorrelation, independent contrasts).

Materials and methods

Species

Captive individuals of 45 carnivore species were filmed while they were feeding (Table 1). The number of individuals per species varied from 1 to 11. For those species for which only one individual was observed (bush dog (*Speothos venaticus*), giant panda, fossa (*Cryptoprocta ferox*), forest genet (*Genetta maculata*), and ocelot (*Leopardus pardalis*)), an attempt was made to corroborate our observations with either documentary footage or published reports. This was not possible for the forest genet, although we did corroborate our observations of the forest genet with observations

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Table 1. The number of individuals observed (*N*), scores for proximal, distal, and total forepaw dexterity (maximum possible score in parentheses), average body mass (BW), degree of arboreality (Arb), relative amount of vertebrate matter in the diet (diet), and references for the arboreality and diet data for the carnivore species examined.

			Forepaw dexterity						
Scientific name	Common name	*N*	Proximal (8)	Distal (13)	Total (21)	BW (kg)	Arb	Diet	References
Family Ailuridae									
Ailurus fulgens	Red panda	7	7	13	20	4.5	3	0	Reid et al. 1991
Family Canidae									
Canis familiaris	Domestic dog	3	2	3	5	30	0	2	Nesbitt 1975
Chrysocyon brachyurus	Maned wolf	4	1	2	3	21.5	0	2	Motta-Junior et al. 1996
Fennecus zerda	Fennec fox	2	1	2	3	1.3	1	2	Bekoff 1975; Nowak 1991
Speothos venaticus	Bush dog	1	1	0	1	6.5	0	3	Langguth 1975; Husson 1978; Peres 1991
Family Felidae									
Acinonyx jubatus	Cheetah	8	1	0	1	53.5	0	3	Kruuk and Turner 1967; Eaton 1972
Caracal caracal	Caracal	3	1	3	4	16	1	3	Grobler 1981
Felis margarita	Sand cat	2	1	3	4	2	1	3	Lay et al. 1970
Felis nigripes	Black-footed cat	2	1	3	4	1.8	1	3	Smithers 1971
Felis silvestris ornatus	Indian desert cat	3	1	3	4	5.5	1	3	Sharma 1979
Leopardus geoffroyi	Geoffroy's cat	2	3	4	7	4.5	1	3	Johnson and Franklin 1991; Yanosky and Mercolli 1994; Branch 1995
Leopardus pardalis	Ocelot	1	8	8	16	13.6	2	3	Mondolfi 1982; Bisbal 1986; Emmons 1988; Sunquist et al. 1989
Lynx canadensis	Canadian lynx	6	1	0	1	11.2	1	3	Nellis et al. 1972; Parker et al. 1983
Panthera leo	Lion	6	6	7	13	156	1	3	Kruuk and Turner 1967; Schaller 1972; Elliott et al. 1977; Ruggiero 1991
Panthera onca	Jaguar	3	8	9	17	105	2	3	Schaller and Vasconcelos 1978; Mondolfi and Hoogersteijn 1982; Rabinowitz and Nottingham 1986; Taber et al. 1997
Panthera pardus	Leopard	3	3	7	13	101	2	3	Kruuk and Turner 1967; Turnbull-Kemp 1967; Karanth and Sunquist 1995; Hart et al. 1996
Panthera tigris	Tiger	4	6	7	13	222.8	1	3	Schaller 1967; Seidensticker and McDougal 1993; Karanth and Sunquist 1995
Panthera uncia	Snow leopard	2	3	4	7	50	1	3	Schaller 1977; Oli et al. 1993
Prionailurus bengalensis	Leopard cat	7	3	9	12	5	2	3	Inoue 1972; Lekagul and McNeely 1977; Rabinowitz 1990
Prionailurus rubiginosa	Rusty-spotted cat	2	1	3	4	2	1	3*	Guggisberg 1975
Prionailurus viverrinus	Indian fishing cat	4	3	10	13	10.9	1	3*	Guggisberg 1975; Lekagul and McNeely 1977
Puma concolor	Cougar	4	7	10	17	81.5	2	3	Hornocker 1970; Ackerman et al. 1984; Taber et al. 1997
Family Herpestidae									
Helogale parvula	Dwarf mongoose	2	1	3	4	0.5	1	1	Rasa 1973, 1977; Kingdon 1977
Suricata suricatta	Meerkat	5	1	3	4	0.9	1	1	Ewer 1963; Doolan and MacDonald 1996
Family Hyaenidae									
Crocuta crocuta	Spotted hyaena	3	1	0	1	63	0	3	Mills 1989, 1990
Hyaena brunnea	Brown hyaena	2	1	0	1	42.3	0	2	Mills and Mills 1978; Mills 1989, 1990

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Family Mustelidae									
Aonyx cinerea	Asian small-clawed otter	2	8	12	20	3	0	0	Kruuk et al. 1994
Eira barbara	Tayra	2	3	5	8	4.5	3	2	Kaufmann and Kaufmann 1965; Brosset 1968; Estes 1980; Bisbal 1986; Konecny 1989; Sunquist et al. 1989
Enhydra lutris	Sea otter	3	5	11	16	38.5	0	0	Tarasoff et al. 1972; Estes 1980; Taylor 1989
Gulo gulo	Wolverine	3	3	5	8	19.5	1	3	Haglund 1968; Hornocker and Hash 1980; C. Long, personal communication
Martes americana	American pine marten	5	3	9	12	1.1	2	2	Leach 1977*a*, 1977*b*; Zielinski et al. 1983; Clark et al. 1987; Thompson and Colgan 1990
Mephitis mephitis	Striped skunk	6	5	10	15	1.6	0	0	Verts 1967; Wade-Smith and Verts 1982
Family Procyonidae									
Nasua narica	Coati	6	6	10	16	4.5	3	2	Kaufmann 1962; McClearn 1992; Gompper 1995
Procyon lotor	Raccoon	7	8	13	21	7	3	1	Whitney and Underwood 1952; Lotze and Anderson 1979; McClearn 1992
Family Ursidae									
Ailuropoda melanoleuca	Giant panda	1	8	13	21	117.5	2	0	Kleiman 1983; Schaller et al. 1989
Tremarctos ornatus	Spectacled bear	2	8	12	20	126	2	0	Peyton 1980; Mondolfi 1989
Ursus americanus	American black bear	5	8	12	20	205	2	0	Herrero 1972; Holcroft and Herrero 1991; Schwartz and Franzmann 1991
Ursus arctos	Brown bear	6	8	12	20	287	1	1	Herrero 1972; Pearson 1975; Servheen 1983; Hamer et al. 1991
Ursus maritimus	Polar bear	4	2	0	2	550	0	3	Russell 1975; Stirling and Archibald 1977; Renous et al. 1998
Ursus thibetanus	Asiatic black bear	2	8	12	20	131.5	2	1	Schaller et al. 1989
Family Viverridae									
Arctictis binturong	Binturong	2	7	7	14	11.5	3	1*	Haglund 1968; Lekagul and McNeely 1977; Wemmer 1977
Cryptoprocta ferox	Fossa	1	7	8	15	9.5	3	3	Wemmer 1977; Köncke and Leonhardt 1986; Laborde 1986*a*, 1986*b*; Goodman et al. 1997; Wright et al. 1997
Genetta genetta	Common genet	2	2	0	2	2	3	3	Taylor 1970, 1972; Delibes et al. 1989; Clevenger 1996
Genetta maculata	Forest genet	1	2	0	2	2	3	3*	Taylor 1970, 1972
Paguma larvata	Masked palm civet	2	1	0	1	4.3	3	1*	Lekagul and McNeely 1977; Wemmer 1977

Note: The scores for proximal, distal, and total dexterity were calculated from the dexterity taxonomy (see Fig. 1). Average body mass were obtained from Nowak (1991), except those for three felids (*Felis margarita*, *Leopardus geoffroyi*, and *Prionailuurs rubiginosa*), which were taken from alternate sources (Guggisberg 1975 and Johnson and Franklin 1991), and the domestic dog (*Canis familiaris*), which was obtained from the average body mass of the three individuals observed. Where extreme variation in body size according to geographical races occurred (i.e., *Panthera pardus*, *Panthera tigris*, *Prionailurus bengalensis*, *Ursus arctos*), the average of all of the subspecies was taken as the species average.

*Studies of actual diet (i.e., stomach or fecal analysis) could not be found for these species and therefore the diet score is an estimate based upon available literature.

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Fig. 1. A tabular representation of the index used to rate forelimb dexterity. The index is divided into three major sections, namely body, forelimb, and forepaw, which correspond to the relative contribution that each body part has in the execution of forelimb movements (Iwaniuk et al. 1999). The three forelimb dexterity scores were calculated as follows: proximal is the sum of the characters under body and forelimb, distal is the sum of the characters under forepaw, and total is the sum of all characters.

BODY	
Number of postures: one, two or three, > three	0, 1, 2

+

FORELIMB	
Limbs cross midline: no/yes	0, 1
Alternate limb-use: no/yes	0, 1
Upper forelimb moves in more than 1 plane: no/yes	0, 1
Upper forelimb rotation: no/yes	0, 1
Lower forelimb rotation: < 90°, 90°-135°, > 135°	0, 1, 2

+

FOREPAW	
Grasping occurs: no/yes	0, 1
Forepaw(s) pick up items: no/yes	0, 1
Unimanual grasping: no/yes	0, 1
"Whole forepaw" grasp: no/yes	0, 1
D2-D3 grasp: no/yes	0, 1
Claw grasp: no/yes	0, 1
"Other" grasp: no/yes	0, 1
Independent digit movements: no/yes	0, 1
Frequency of manipulation: none/rare, up to 60%, > 60%	0, 1, 2
Items swapped between forepaws: no/yes	0, 1
Items rolled between forepaws: no/yes	0, 1
Distal digits used in manipulation: no/yes	0, 1
Forepaws pull away from one another: no/yes	0, 1
"Other" digit/forepaw movements: no/yes	0, 1

of identical behaviours in common genets (*Genetta genetta*) and published reports of *Genetta tigrina* and *Genetta servalina* (Wemmer 1977).

Film footage of all of the species was subjected to frame by frame analysis and each species was coded according to a forelimb dexterity index (see the following section).

Behavioural coding

Given the problem inherent in previous dexterity indices (discussed earlier), we utilized an index that not only combines proximal and distal contributions of the forelimb, but also allows their separation. The rationale for this index has been discussed elsewhere (Iwaniuk et al. 1999). Each species received a numerical score for each of the categories within the index (Fig. 1). The scores were then tallied to produce separate proximal and distal scores and a total score (Table 1).

To assess the forelimb dexterity of the 45 species, we filmed while they were feeding. Feeding was chosen to assess forelimb dexterity for two reasons: feeding behaviour occurs relatively frequently in captivity and is readily replicated, and previous studies have shown that such behaviours are appropriate for assessing forelimb dexterity (Whishaw and Pellis 1990; Whishaw et al. 1993, 1998; Iwaniuk et al. 1998; Iwaniuk and Whishaw 1999). Since feeding is a subset of the behaviours that involve forelimb usage, an attempt was made to corroborate observations of forelimb usage in feeding with that of other behaviours, for each of the species examined. Forelimb use during both locomotion and grooming was significantly less variable than that during feeding. In two species, the cheetah and the masked palm civet (*Paguma larvata*), patterns of predatory behaviour exhibited slightly different variability in forelimb use (personal observation; Eaton 1972; Eisenberg and Leyhausen 1972), but the absence of published reports and (or) documentary footage of predation for most of the species pre-

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cluded the inclusion of predation for all species. Even so, the dexterity index for these two species would have only been increased by a maximum of two or three points and was not expected to significantly affect the significance of the results when altered. Furthermore, hypotheses concerning the evolution of forelimb dexterity have themselves focussed upon feeding behaviour (Gittleman 1986*a*; Napier 1980), and therefore feeding seemed an appropriate behaviour to examine.

To test the hypotheses that arboreality and vertebrate predation are both positively correlated with forelimb dexterity, a coding system was required for these behaviours. Previous studies of locomotion and diet in carnivores have chosen to use either unordered categories (Van Valkenburgh 1985, 1987; Gittleman 1986*a*, 1986*b*) or exact percentages (Gittleman and Harvey 1982). Behaviours rarely occur, however, as mutually exclusive categories, and the use of precise proportions or percentages can be subject to change depending upon seasonal, geographical, and methodological differences. We therefore chose to use indices that were based upon specific proportions–percentages in the literature, but were slotted into ordered categories that were gross enough so that seasonal and other minor fluctuations would not significantly affect species' coding. Thus, arboreality was coded as follows: 0, never climb trees; 1, capable of climbing, but do so rarely (i.e., scansorial); 2, good climbing skills and climb frequently (i.e., semiarboreal); and 3, excellent climbing skills and capable of controlled, head-first descent (i.e., arboreal) (Table 1). Similarly, vertebrate predation was divided into four ordered categories contingent upon the estimated biomass of vertebrates in the diet: 0, <25%; 1, 25–50%; 2, 50–75%; and 3, >75% (Table 1). Where specific dietary data were not available, an estimate of the dietary index, based upon available information, was used. For example, there are no known detailed studies of binturong (*Arctictis binturong*) diets in the wild, so the diet described in Lekagul and McNeely (1977) was used to score the binturong as a 1 (Table 1). Although this may reduce the reliability of this index, it was only necessary for four out of the 45 species (Table 1).

Statistical analysis

To assess the relationships between body size, phylogeny, arboreality, and vertebrate predation and forelimb dexterity, it was necessary to perform four levels of analysis. First, body size effects were controlled for. The dexterity scores and body mass (from Nowak 1991) were logarithm transformed and regressed against each other using least-squares linear regression (LaBarbera 1989). If a significant result was found, the residuals from that analysis were used in the subsequent analyses of phylogeny and behaviour.

The second method, autocorrelation, was used to estimate the amount of interspecific variation that can be attributed to phylogeny. This method uses autoregression techniques to partition the total amount of variation in a trait into a phylogenetic and an environmental (i.e., adaptive) component (Cheverud et al. 1985). A Moran's *I* statistic was used to determine the phylogenetic distance at which autocorrelation should be performed (Gittleman and Kot 1990; Gittleman and Luh 1992). Through the use of a matrix that encodes the phylogenetic relatedness between the species of interest, predicted values of the trait can be calculated and correlated with the observed values to yield the amount of variation due to phylogeny.

To perform this analysis, the phylogenetic autocorrelation (PA) program was used (Luh et al. 1995). Although taxonomic ranks can be used in autocorrelation (Gittleman and Luh 1992), the use of branch lengths is likely to produce more accurate and meaningful results simply due to the finer definition of interspecies relatedness provided by known branch lengths (J.L. Gittleman, personal communication; Gittleman and Luh 1992). Branch lengths were obtained from the tables provided in Bininda-Edmonds et al. (1999).

Lastly, to examine the relationship between arboreality and vertebrate predation and forelimb dexterity, we performed ANOVAs of the three dexterity scores relative to the arboreality and diet scores. However, since the use of species as independent data points violates the assumptions of independence in conventional statistics (Harvey and Pagel 1991), we also employed Felsenstein's (1985) method of independent contrasts. Simulations of different comparative techniques have shown that independent contrasts is the most robust method (Diaz-Uriarte and Garland 1996, 1998; Garland et al. 1999) and yields the lowest type I error rates (Martins and Garland 1991). Since branch lengths were known, a gradual model of evolutionary change was used, as it has been shown to be the most robust form of independent-contrast analysis (Martins and Garland 1991).

These methods requires a phylogenetic hypothesis, which was taken from Bininda-Edmonds et al. (1999) (Fig. 2). For the purposes of comparative analysis, the domestic dog (*Canis familiaris*) was treated as equivalent to the wolf (*Canis lupus*) in the phylogeny of Bininda-Edmonds et al. Resolution was not possible in Bininda-Edmonds et al. for the maned wolf (*Chrysocyon brachyurus*) and the bush dog, and therefore their relationship to the other canids was taken from Wayne et al. (1997). Resolved relationships among all of the species were necessary because polytomies are unlikely to be representative of speciation events and independent-contrasts analysis is more reliable when all of the speciation events are unambiguously resolved (Purvis and Rambaut 1995). Furthermore, the likelihood that changes in the terminal branches will significantly affect the results is negligible (Price 1997).

The phylogeny, branch lengths, and data set were entered into the comparative analysis of independent contrasts (CAIC) program (Purvis and Rambaut 1995). All of the traits were considered to be continuous. Once the contrasts were calculated, proper standardization of the data was tested by regressing the absolute value of the contrasts against the square root of the branch lengths (Garland et al. 1993). None of the regressions were significant (Table 2), indicating that the data were properly standardized. Normality of the data was then tested with a Shapiro–Wilk test for each variable (JMP version 3.2, SAS Institute Inc., Cary, N.C.). All of the dependent variables (i.e., dexterity scores) were normally distributed, but the independent variables were not ($W_{\text{arboreality}} = 0.86$, $P < 0.01$; $W_{\text{diet}} = 0.75$, $P < 0.01$). Logarithm, square-root, and cube-root transformations were not successful in normalizing the data, so the untransformed contrasts were used in subsequent regression analyses. Although the lack of a normal distribution in the independent variable can be problematic in regressions, the normal distribution of the dependent variable is a more crucial assumption (Daniel 1991). Lastly, the dexterity contrasts were regressed against the arboreality and vertebrate predation contrasts through the origin (Garland et al. 1992) using reduced major axis regression formulae, and an *F* test was used to assess significance. Despite the possibility of nonparametrics being affected by the repeated zeros present in independent contrasts (Garland et al. 1993), Spearman-rank correlations were also performed to account for the non-normal distribution of the independent variables.

Lastly, to ascertain the pattern of evolutionary change in forelimb dexterity, we mapped the three dexterity scores onto the phylogeny of Bininda-Edmonds et al. (1999) using MacClade (Maddison and Maddison 1992). Although the dexterity index is not continuous in the true sense (i.e., cannot have a total dexterity of 6.71), the presence of large integers (i.e., above 10) precluded the use of the discrete character algorithms, and therefore the dexterity scores were treated as continuous variables. Two algorithms were chosen to determine the ancestral state for the "basal" carnivore node and that for the basal caniform and feliform nodes. The first of these uses a linear parsimony model to minimize the sum of the evolutionary changes occurring across the entire tree (Martins

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Fig. 2. The phylogeny of the 45 species examined taken from Bininda-Edmonds et al. (1999). Note that the domestic dog (*Canis familiaris*) occupies the position taken by the wolf (*Canis lupus*) in Bininda-Edmonds et al. and that resolution of the maned wolf (*Chrysocyon brachyurus*) and the bush dog (*Speothos venaticus*) is taken from Wayne et al. (1997).

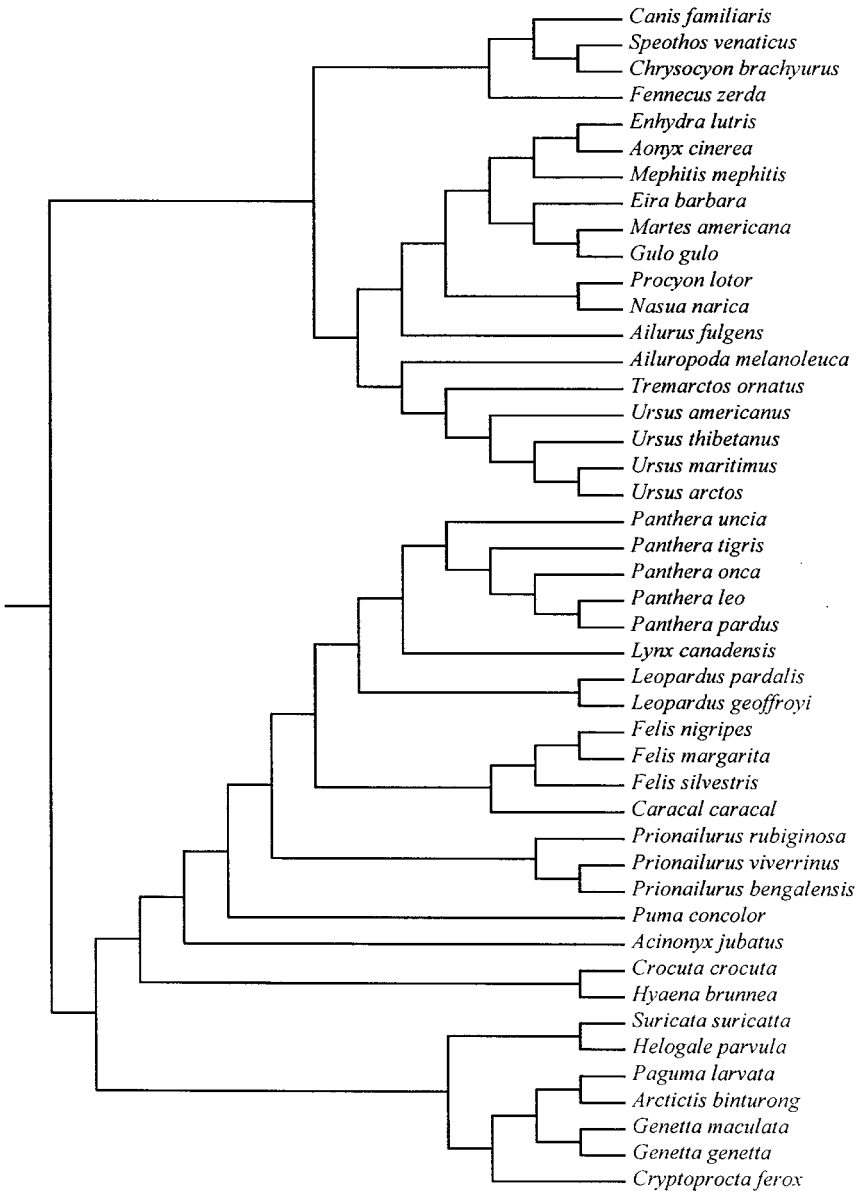


Table 2. Results from the regression analysis of the absolute value of the contrasts against the square root of the branch lengths of the phylogeny.

Contrast	$F_{[1,42]}$	P
Proximal dexterity	3.506	0.0681
Distal dexterity	0.284	0.5970
Total dexterity	0.308	0.5821
Arboreality	3.577	0.0655
Vertebrate predation	1.158	0.2880

and Hansen 1996). Although this is the most commonly used method of ancestral state reconstruction (Martins and Lamont 1998), it can be problematic when the rate of evolution is not slow (Martins and Hansen 1996) and often yields multiple solutions (Swofford and Maddison 1987). A second method was therefore implemented. The sum of squared changes parsimony model minimizes the sum of the squared changes across the entire tree (Martins and Hansen 1996). This model not only handles different rates of evolutionary change (e.g., Brownian motion), but also yields a single value.

Results

Body size

All three dexterity indices were positively correlated with body size, but only the relationship with proximal dexterity index was significant ($F_{[1,43]} = 9.49$, $P < 0.05$). This result suggests that with increasing body size, there is a significant increase in proximal forelimb dexterity. Given that this analysis does not incorporate phylogeny, and that the result could be affected by nonindependence of data, the independent-contrasts approach was also applied to the analysis of body

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Fig. 3. The results of the Moran's *I* test on the three dexterity scores. The *x* axis represents the divergence time of the nodes of the phylogeny in millions of years (Ma) and is based on the tables provided in Bininda-Edmonds et al. (1999). The *y* axis represents the *Z* scores calculated by the PA program (Luh et al. 1995) for the nodes. Since all three dexterity scores possess *Z* scores greater than 1.96 at one or more phylogenetic distances, they exhibit phylogenetic autocorrelation.

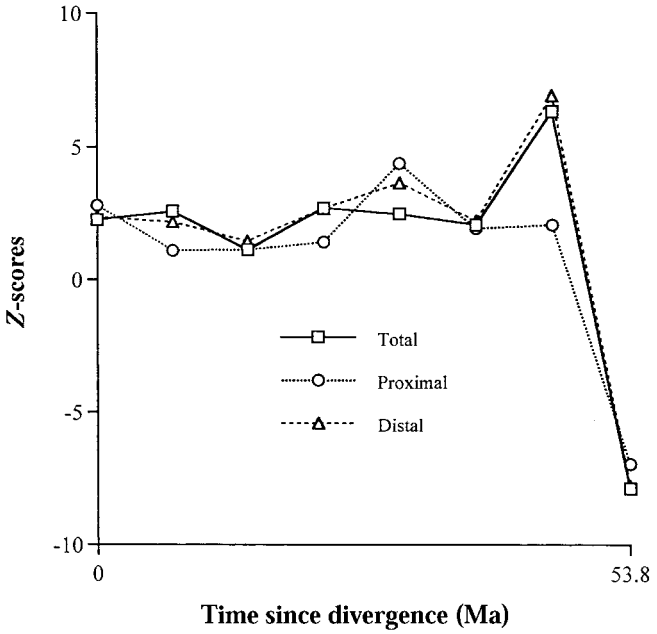


Table 3. Autocorrelation results for the three dexterity indices in which α is the statistical power of the weighting matrix, ρ is the autocorrelation coefficient, and r^2 is the amount of the total variation accounted for by phylogeny (i.e., Pearson's *r*).

	α	ρ	r^2
Proximal dexterity	9.30	0.37	0.2662
Distal dexterity	7.90	0.27	0.1443
Total dexterity	7.35	0.27	0.1456

size. Once phylogenetic relatedness was incorporated, the relationship between proximal dexterity and body size became nonsignificant ($F_{[1,42]}$ = 0.0004, P = 0.99). Distal and total dexterity scores and body size remained nonsignificant.

Phylogeny

The results of the Moran's *I* test indicated that phylogenetic distance is significantly correlated with all three dexterity scores (Fig. 3). Subsequent autocorrelation determined that the amount of interspecific variation explained by phylogenetic relatedness varied between the dexterity scores (Table 3).

Arboreality

ANOVAs of the species as independent data points indicated that there were significant differences in all three dexterity scores with respect to arboreality (Table 4). Subsequent Tukey–Kramer tests indicated that species that scored 0 or 1 had significantly lower dexterity scores than those that scored 2. Thus, nonarboreal and scansorial species had significantly lower dexterity scores than semiarboreal species.

Table 4. The results of the ANOVAs of the dexterity scores relative to the arboreality and diet scores.

Dexterity score	$F_{[3,41]}$	P
Arboreality		
Proximal	7.70	<0.01
Distal	5.55	<0.01
Total	6.74	<0.01
Diet		
Proximal	2.04	0.12
Distal	3.44	0.03
Total	2.95	0.04

Table 5. The slopes (β) and r^2 values for the regression lines of the three dexterity indices against arboreality and vertebrate predation (diet) with and without the rogue brown bear – polar bear contrast.

	With		Without	
	β	r^2	β	r^2
Arboreality				
Proximal	6.7824	0.2380*	9.5813	0.1039
Distal	0.8352	0.1681	0.3872	0.0967
Total	1.3280	0.1774	0.6628	0.1058
Diet				
Proximal	–4.4582	0.5236**	–2.9185	0.4763**
Distal	–4.0510	0.7003**	–2.6420	0.2778**
Total	–6.0740	0.6568**	–3.9570	0.2326*

Note: *, P < 0.05; **, P < 0.01.

This was not, however, corroborated by regression analyses of the independent contrasts. Only the regression of the proximal dexterity against arboreality yielded a significant relationship (Table 5). The presence of an outlier, the contrast between the brown bear (*Ursus arctos*) and the polar bear (*Ursus maritimus*), appeared to alter both the correlation coefficient (r^2) and the slope of the regression line and was well outside of the statistically acceptable range of two standard deviations (contrast value = average of 4.2 standard deviations from the regression lines) (Fig. 4*a*). Removal of the rogue point resulted in no significant correlations being found (Table 5). Thus with an increase in arboreality, there is no correlated change in proximal, distal, or total dexterity. This was corroborated by the results of Spearman-rank correlations of the contrasts ($Z_{proximal}$ = 1.65, n = 44, P > 0.05; Z_{distal} = 1.55, n = 44, P > 0.10; Z_{total} = 1.64, n = 44, P > 0.10).

Vertebrate predation

ANOVAs of the species as independent data points indicated that there were no significant differences in proximal dexterity across the diet scores, but there were significant differences for both distal and total dexterity (Table 4). Subsequent Tukey–Kramer tests indicated that species that scored 0 had significantly higher dexterity scores than those that scored 3. That is, species that relied less heavily upon vertebrates as part of their diet had higher degrees of fore-

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limb dexterity than those that were primarily vertebrate predators.

This was supported by regression analyses of the independent contrasts (Table 5). As with the arboreality regressions, the brown bear – polar bear contrast was a rogue point for the comparison of vertebrate predation and the dexterity scores (Fig. 4*b*). Despite the changes in slope and r^2 when the rogue point was included, the significance of the results was the same: all three dexterity scores were significantly correlated with vertebrate predation (Table 5). Contrary to the prediction, however, dexterity was found to decrease with increasing degrees of vertebrate predation. Thus, species that rely more upon vertebrates have lower degrees of dexterity in both proximal and distal components. This was only partially supported by Spearman-rank correlations ($Z_{proximal} = -1.73$, $n = 44$, $P > 0.05$; $Z_{distal} = -2.31$, $n = 44$, $P < 0.05$; $Z_{total} = -1.94$, $n = 44$, $P < 0.05$).

Character mapping

Linear parsimony reconstruction of the proximal, distal, and total dexterity scores indicated low, unresolved scores for the basal caniform, feliform, and carnivore nodes (Table 6). Sum of squared changes parsimony yielded higher, single values for each of the nodes (Table 6; Fig. 5).

Discussion

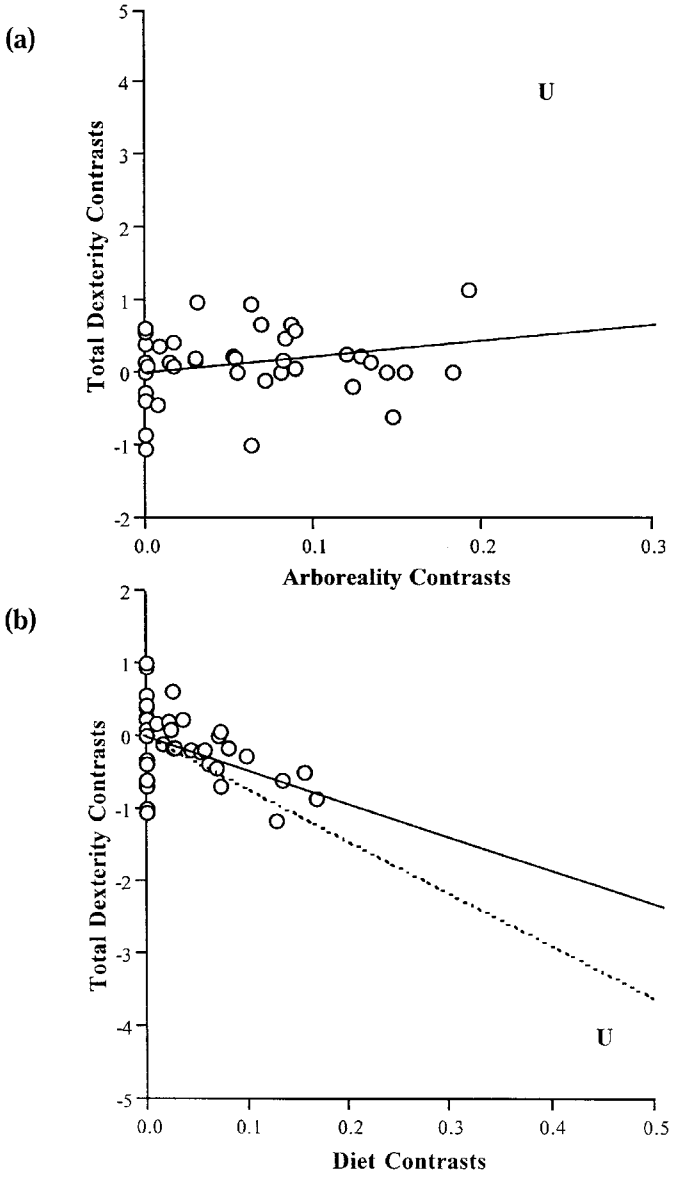
Both phylogeny and vertebrate predation explained a significant amount of the interspecific variation in forelimb dexterity in carnivores (Table 5). The amount of variation did, however, vary within both of the independent variables examined, thus supporting our view that the components of the forelimb can be elaborated upon separately in response to different functional demands.

Caveats

It could be argued that in not examining a greater number of carnivore species we may be missing some interspecific variation that could significantly affect our results. Although we only examined 19% of all of the fissiped carnivores, preliminary observations of 15 additional species (four canids, one felid, seven mustelids, one ursid, and one viverrid) increase the likelihood that the diversity of forelimb dexterity presented herein is representative of the taxon.

A second potential source of error may be the phylogeny employed. The phylogeny of Bininda-Edmonds et al. (1999) is a "supertree" resulting from the analyses of various phylogenetic data sets. Morphological, molecular, and chromosomal data are therefore combined to form a larger data set that is then analyzed by a method known as matrix representation with parsimony analyses (MRP) (Baum 1992; Ragan 1992). Although this method often yields trees that are well resolved (e.g., Bininda-Edmonds et al. 1999; Purvis and Rambaut 1995), there are a number of problems inherent in this method, such as being node-based rather than tree-based in determining relationships and placing taxa in positions that they would not occupy with the use of other total-evidence analyses (reviewed in Bininda-Edmonds and Bryant 1998). Thus, the position of some species on the carnivore phylogeny of Bininda-Edmonds et al. (1999) may be questionable. In the absence of other suitably detailed carnivore phylogenies, however, the phylogeny used herein was the best one currently available. Furthermore, relatively minor alterations in the position of individual species in relatively large data sets appear to have little influence on the significance of independent-contrasts regressions (Price 1997; A.N. Iwaniuk, unpublished data).

Fig. 4. A plot of the independent contrasts of the total dexterity scores against arboreality (*a*) and diet (*b*). The rogue contrast between the brown bear (*Ursus arctos*) and the polar bear (*Ursus maritimus*) is indicated by U. Where differences in the slope of the regression line occurred as a result of the presence of this rogue contrast, the broken line refers to the regression line including the rogue point and the solid line refers to the regression line excluding the rogue point. Note that similar relationships also occurred for the other two dexterity scores (see Table 4).



Body size

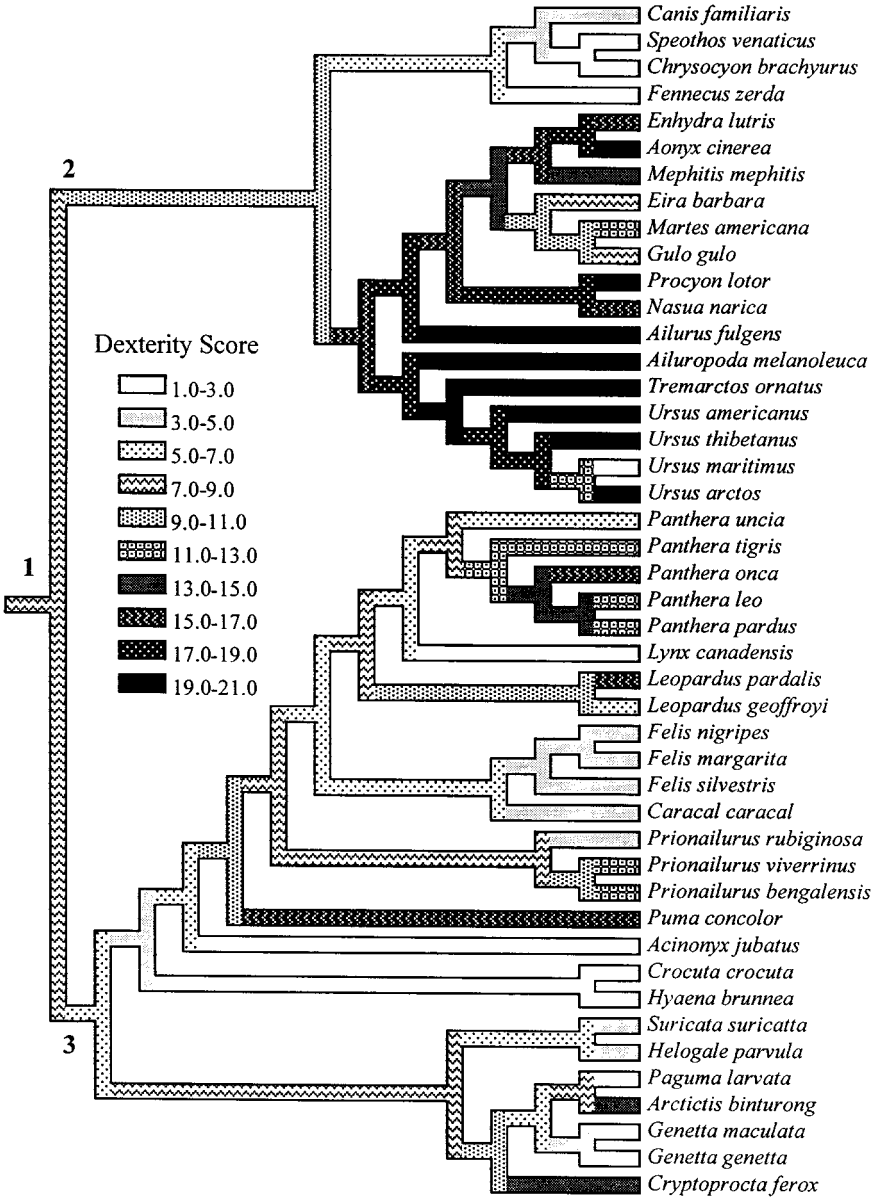
Overall body size plays a significant part in most aspects of an organism's biology (Schmidt-Nielsen 1984; LaBarbera 1989; Damuth and MacFadden 1990). Despite this fact, none

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Table 6. The ancestral states of the three dexterity scores, using two models of reconstruction, at the three basal nodes of the carnivore phylogeny (see Fig. 5).

Model and dexterity score	Basal carnivore	Basal caniform	Basal feliform
Linear parsimony			
Proximal	3–4	3–4	3–4
Distal	2–3	2–3	2–3
Total	3–4	3–4	3–4
Sum of squared changes Parsimony			
Proximal	3.3	3.7	2.9
Distal	4.9	5.9	3.8
Total	8.2	9.6	6.7

Fig. 5. The reconstruction of the ancestral states of the total dexterity score using sum of squared changes parsimony. The three numbered nodes are as follows: 1, basal carnivore; 2, basal caniform; 3, basal feliform. The exact values for these three nodes and the other two dexterity scores are given in Table 6.



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of the dexterity scores were found to be significantly correlated with body size. This would appear to be related to a strong phylogenetic component in the variation of both forelimb dexterity (see below) and body size (Gittleman 1986*b*).

A second characteristic of body size that may have obscured any relationship with forelimb dexterity is that body size is the result of the large number of selective forces that act independently of forelimb dexterity (Schmidt-Nielsen 1984; LaBarbera 1989; Damuth and MacFadden 1990). Therefore, in most species, the interactions between body size and many different factors are likely to be relatively stronger than any possible relationship between body size and dexterity. Thus, large species, depending upon their phylogenetic history and diet (see the following sections), are equally capable of evolving high degrees of forelimb dexterity as smaller species.

Phylogeny

The phylogenetic, or inherited, portion of variation in a trait can have a significant effect on the evolution of behaviours (Harvey and Pagel 1991). Phylogenetic relatedness accounted for a significant amount of the interspecific variation in all three dexterity scores. This suggests that there are some phylogenetic constraints that may shape the evolution of forelimb dexterity in mammals. How and why this occurs is difficult to resolve. It may occur due to phylogenetic time lags, developmental constraints, phylogenetic niche conservatism, or a myriad of other possibilities (Harvey and Pagel 1991).

Studies of forelimb dexterity in other mammalian taxa do support a significant phylogenetic effect on forelimb use. For example, Whishaw et al. (1998), from an examination of forelimb use and food handling in 10 different rodent species, concluded that not only do rodents appear to share a common structural pattern in their forelimb movements during feeding, but also that species that are more closely related to one another tend to be more similar in their forelimb use than those that are distantly related. Similarly, in the study by Bishop (1964) of forelimb use in primates, species that belonged to the same families had a tendency to exhibit forelimb use patterns more similar to those of one another than to those of species in other families. For example, lorises (family Loridae) possess a common set of forelimb movements,which they use to capture prey, whereas bushbabies (family Galagidae) execute a different set of forelimb movements. Corroborating evidence is found more recently within species of marmosets and tamarins (Singer and Schwibbe 1999). Thus, there is a trend amongst mammalian taxa to share a common set of forelimb movements, and dexterity, which is supported by the present results.

Arboreality

Arboreal locomotion, the movement of an animal through the trees, has been regarded as a key factor promoting complex forelimb use (Cartmill 1985; Nudo and Masterton 1990). Although the analysis of the "raw" data indicates that there is a relationship between climbing skill and forelimb dexterity (Table 4), the more robust independent-contrasts results indicate that there is no significant relationship (Table 5). This may be, in part, due to the lack of extremes of arboreal specializations in the carnivores examined, but it appears to indicate that grasping forepaws may not be an essential part of locomoting through the trees.

Arboreal locomotion can, in fact, be facilitated by a number of other morphological adaptations apart from grasping forepaws. For example, claws may be used to secure purchase on tree trunks. Among the species examined, the American pine marten (*Martes americana*) possesses relatively mediocre forelimb dexterity (Table 1), yet this species is capable of rapidly pursuing squirrels through the trees (Clark et al. 1987). Rather than using grasping forepaws, the marten has needle-sharp claws that serve the dual purposes of retaining contact with the substrate during arboreal locomotion and facilitating effective prey capture. The use of claws in arboreal locomotion also occurs in members of other taxa such as the marsupial phascogales (*Phascogale* spp.) and North American tree squirrels (*Sciurus* spp. and *Tamiasciurus* spp.). Other mechanisms may also be used that do not rely upon grasping such as adhesion (reptiles and amphibians) and suction (some bats) (Cartmill 1985). Thus, arboreal locomotion does not necessarily require grasping forepaws and, consequently, does not appear to be directly correlated with forelimb dexterity.

Vertebrate predation

Predatory behaviour, specifically the capture of vertebrate prey, has been considered to be a factor promoting forelimb dexterity (Hopson 1977; Gittleman 1986*a*). That is, species that capture vertebrates possess higher degrees of forelimb dexterity than those that do not. Contrary to this theory, our results demonstrate that, although a significant correlation does exist between forelimb dexterity and vertebrate predation, it is negative rather than positive. This is borne out not only in the regression analysis of independent contrasts (Table 5), but also in the raw analysis (Table 4). Thus, species that feed exclusively upon vertebrates tend to have a lower degree of forelimb dexterity than species feeding on other items.

Although predatory species, in general, do require considerable skill in the use of their forelimbs to successfully capture and manipulate prey, they also require the ability to chase down prey. This is evident even in ambush predators that may have to pursue potential prey in short chases prior to capture (A.N. Iwaniuk, personal observation). This locomotor requirement may have placed limitations on the flexibility of the forelimbs, despite their use as prey-capture organs. An examination of detailed descriptions of predatory behaviour (Eisenberg and Leyhausen 1972; Pellis and Officer 1987; Ben-David et al. 1991) indicates that, although the manipulative movements are rapid, precise and skilled movements of the distal components are relatively unimportant. Rather, it is the proper placement of the teeth that is of utmost importance in subduing prey. In fact, in some species, complex prey capture can occur in species with relatively poor proximal and distal dexterity (e.g., opossums (*Monodelphis domestica*), Ivanco et al. 1996). Thus a combination of locomotor requirements and a lack of intensive selection for high degrees of dexterity in predatory species may limit the degree of dexterity exhibited in both proximal and distal components.

In contrast with predatory species, omnivorous species do not require locomotor adaptations for pursuing prey, since

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they are generally opportunistic. Also, by definition, omnivores feed upon a wide variety of foods, which can range considerably in shape, size, and texture. To feed upon such a wide variety of foods and manipulate them effectively probably requires a greater degree of distal dexterity than prey capture (Ivanco et al. 1996; Iwaniuk et al. 1998). Furthermore, the ability to mechanically process foodstuffs could be improved by using the forelimb to manipulate and (or) process items so that maximal processing efficiency can occur (Iwaniuk 1996). Thus, the comparative lack of locomotor restrictions and possible selection for higher degrees of forelimb dexterity are likely to result in higher degrees of forelimb dexterity in omnivores.

It should be noted, however, that some of the species given a low vertebrate predation score, such as the crustacean specialist, the small-clawed otter (*Aonyx cinerea*), and the herbivorous giant panda, are not true omnivores. These species are, in fact, dietary specialists in much the same way as vertebrate predators. The key difference in these two species is that their dietary specializations require high degrees of forelimb dexterity. For example, in the small-clawed otter, crabs and other aquatic invertebrates are taken underwater from fissures and under rocks (Kruuk et al. 1994). This kind of extractive foraging is likely to promote forelimb dexterity, as it has in other extractive foragers (e.g., capuchin monkeys (*Cebus* spp.)), because finely tuned movements are an advantage in foraging for and grasping prey underneath and within different objects. The giant panda, however, feeds mainly upon bamboo (Schaller et al. 1989), which does not require extractive foraging techniques. By being able to manipulate bamboo culms, the giant panda may be able to mechanically process the bamboo more effectively and therefore consume more bamboo per sitting. Support for this theory is found in the similar manipulative behaviours that have been reported for gentle lemurs (*Hapalemur griseus*) feeding on bamboo (Stafford et al. 1993). Thus, in general, forelimb dexterity decreases with vertebrate predation, which is suggestive of a correlation between omnivory and dexterity, but specific dietary specializations can also promote forelimb dexterity.

Ancestral states

The reconstruction of the dexterity scores at ancestral nodes indicates that the ancestor of each of two major lineages and that of the entire order had moderate to low dexterity. The examination of the pattern of evolutionary changes across lineages further indicates that, in some cases, particular changes in forelimb dexterity characterize entire lineages. For example, within the canids, forelimb dexterity has decreased, but it has increased within the ursids (Fig. 5). In some lineages, however, there are species that are drastically different in their forelimb dexterity than their closest relatives. One such example is the polar bear. In comparison to the rest of the ursids, the polar bear is unique in that it has lost the majority of its forelimb dexterity. This can best be explained by the niche occupied by the polar bear. Unlike other ursids, the polar bear is almost entirely carnivorous and ranges over tundra regions. To move efficiently over large distances in search of prey (up to 300 km per season; Stirling et al. 1978; Schweinsburg and Lee 1982), the polar bear also possesses a suite of morphological and behavioural adaptations that set it apart from its closest relatives, such as longer, stiffer limbs, a longer trunk, and dense mats of fur covering the paws (Renous et al. 1998). This strong reliance upon efficient locomotion appears to have been at the expense of forelimb dexterity, as the polar bear possesses significantly lower dexterity scores than other bears. Thus, in the polar bear, the requirements of locomotion seem to have superseded that of forelimb dexterity.

Similar arguments can be made for species differences within other clades, the point being that forelimb dexterity may also be an adaptive response to selection pressures specific to a species. As such, changes in forelimb dexterity may also be viewed as an integral part of invading specific niches. For example, there is a tendency for riparian species (those species inhabiting watercourses) to have relatively high degrees of forelimb dexterity (e.g., the raccoon (*Procyon lotor*) and the small-clawed otter). Although a causal mechanism cannot be inferred from correlational analyses, the relationship between riparian foraging and forelimb dexterity may be indicative of a functional response to an overall niche. To determine whether niche metrics can be related to forelimb dexterity will, however, require study of a broader range of species and more detailed information on the dietary habits of the species examined.

Conclusions

Body size and climbing ability are not correlated with the forelimb dexterity of carnivores, but phylogeny and vertebrate predation are. A significant part of the variation, within carnivores at least, is likely due to species-specific selection pressures. This would suggest that the evolution of complex motor behaviours may be adaptive and therefore may play a significant role in the invasion of some ecological niches. Whether the relationships presented herein will hold true for other mammalian taxa requires analysis. It may be instructive to conduct similar forms of analyses in other taxa known to possess species with variable levels of forelimb dexterity, particularly the Rodentia, Marsupialia, and Primates. General correlational patterns across different lineages may not only demonstrate the importance of forelimb dexterity in particular niches, but also provide useful insight into the morphological bases underlying both high and low forelimb dexterity.

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sin), Mountain View Breeding Farms Conservation and Breeding Center (Langley, British Columbia), Oaklawn Farm Zoo (Aylesford, Nova Scotia), Paris Zoo (Paris, France), Shubenacadie Wildlife Park (Shubenacadie, Nova Scotia), Vancouver Aquarium (Vancouver, British Columbia), and the Valley Zoo (Edmonton, Alberta). This study partially fulfilled the requirements of a M.Sc. thesis by the primary author. Funding for this study was provided by grants to Ian Q. Whishaw and a postgraduate scholarship to Andrew N. Iwaniuk from the Natural Sciences and Engineering Research Council of Canada.

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